

Requirement Analysis Phase

Data Flow Diagrams and User Stories

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Team ID	SWTID-2026-7059
Project Name	FitBuddy
Maximum Marks	4 Marks

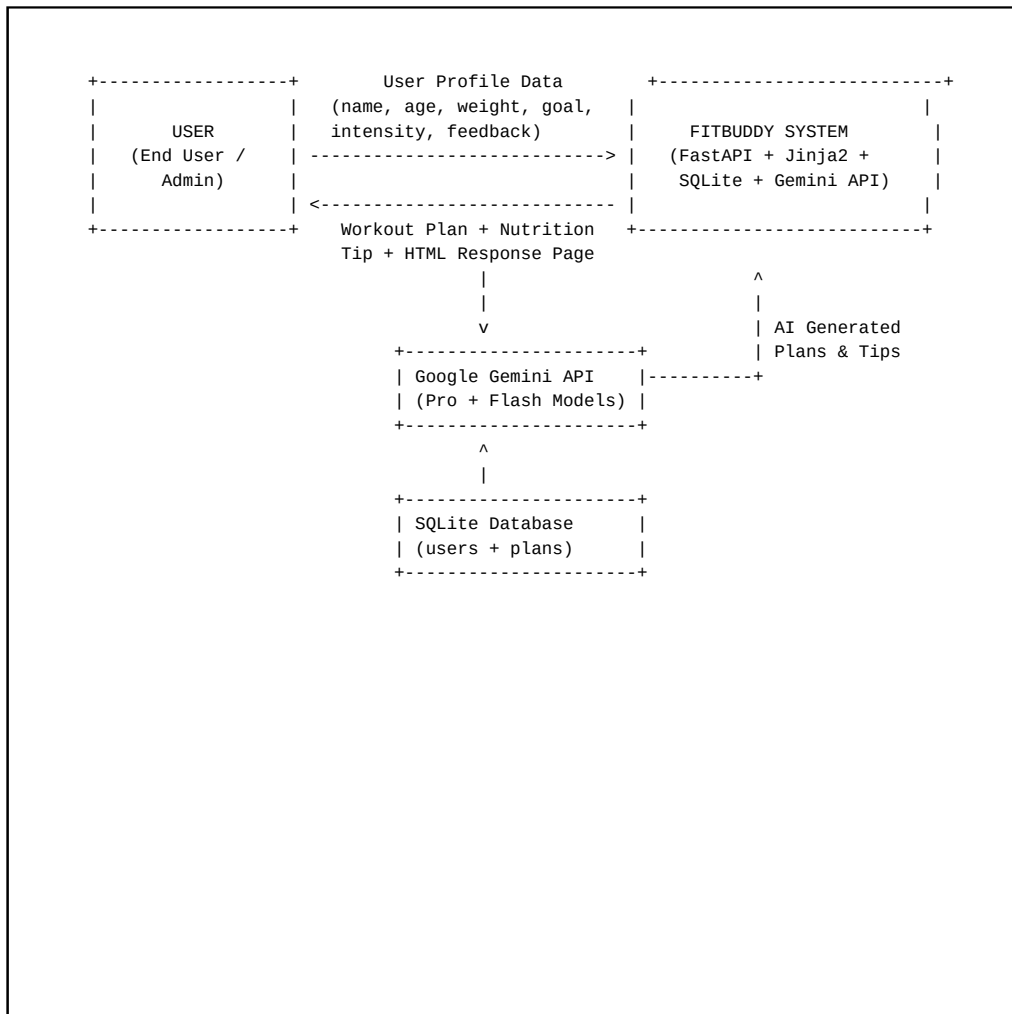
1. Introduction

This document presents the Data Flow Diagrams (DFD) and User Stories for **FitBuddy** — an AI-powered personalized fitness coaching web application. The DFDs illustrate how data moves through the system from user input to AI-generated output and database storage. The User Stories capture the functional requirements from the end-user's perspective.

2. Data Flow Diagram – Level 0 (Context Diagram)

The Level 0 DFD shows FitBuddy as a single process with its external entities and data flows.

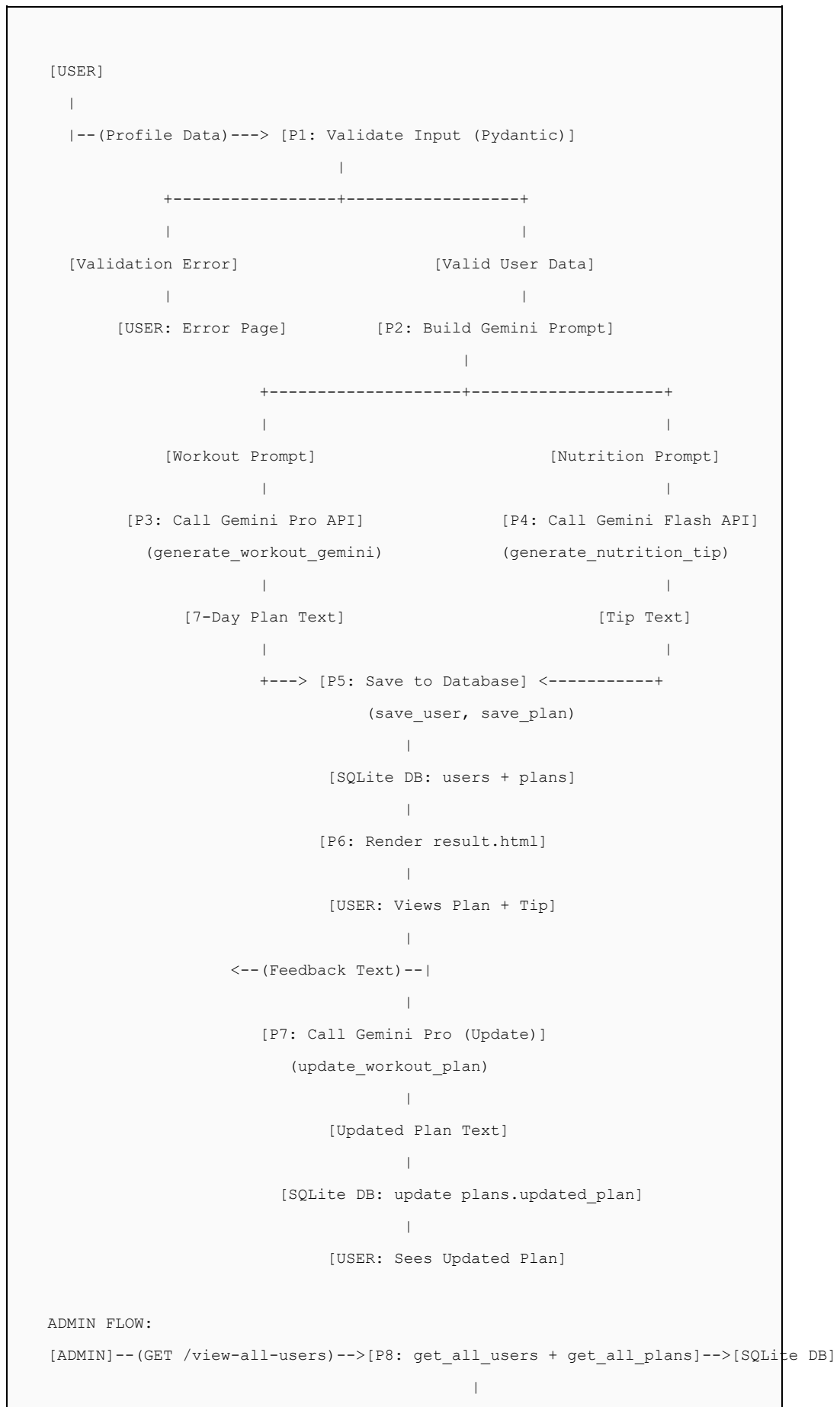
Figure 1: FitBuddy System Context Diagram (DFD Level 0)



3. Data Flow Diagram – Level 1 (Process Decomposition)

The Level 1 DFD decomposes the FitBuddy system into its core processes.

Figure 2: FitBuddy Process Decomposition (DFD Level 1)



[Admin Dashboard Page]

[ADMIN] -- (POST /delete-user/{id}) --> [P9: delete_user] --> [SQLite DB]

4. Data Store Definitions

Table	Column	Data Type	Description
users	id	Integer (PK)	User-supplied numeric ID (unique identifier, not auto-incremented)
	name	String(100)	User's full name
	age	Integer	User's age (validated: 10 to 100)
	weight	Float	User's body weight in kilograms (validated: 20 to 400)
	goal	String(100)	Fitness goal, e.g. "weight loss", "muscle gain", "flexibility"
	intensity	String(20)	Workout intensity level: low / medium / high
	schedule	Integer	Number of days in the plan (default: 7)
plans	id	Integer (PK)	Auto-incremented plan ID
	user_id	Integer (FK)	Foreign key linking to users.id (one-to-one relationship, cascade delete)
	original_plan	Text	Full AI-generated 7-day workout plan from Gemini Pro (not null)
	updated_plan	Text (nullable)	Feedback-revised plan from Gemini Pro; null if no feedback has been submitted

5. User Stories

The following user stories define the functional requirements from the end-user perspective, with acceptance criteria for each story.

US-01 | Generate Personalized Workout Plan | Story Points: 8 | Priority: High

*As a **fitness enthusiast**, I want to enter my name, age, weight, fitness goal, and workout intensity so that I can receive a personalized AI-generated 7-day workout plan.*

Acceptance Criteria:

- User can fill a form with: Name, User ID, Age (10–100), Weight (20–400 kg), Goal (free text), Intensity (Low / Medium / High)
- All fields are validated; invalid inputs display inline error messages
- On submission, Gemini Pro generates a structured 7-day plan
- Plan includes warm-up, main workout (exercises, sets & reps), and cooldown for each day
- At least one rest or active recovery day is included in the plan
- Plan is displayed on the result page and saved to the SQLite database

US-02 | Receive AI Nutrition Tip | Story Points: 3 | Priority: Medium

*As a **user viewing my workout plan**, I want to see a nutrition or recovery tip relevant to my fitness goal so that I can support my training with proper diet advice.*

Acceptance Criteria:

- A nutrition/recovery tip is displayed alongside the workout plan on the result page
- The tip is generated by Gemini Flash based on the user's specific fitness goal
- If Gemini Flash is unavailable, a pre-defined fallback tip is shown (no error displayed)
- Tip is 2–3 sentences in plain text with no markdown formatting

US-03 | Refine Plan with Feedback | Story Points: 5 | Priority: High

*As a **user who has received a workout plan**, I want to submit feedback (e.g., "add more cardio", "less leg days") so that the AI updates my plan to better match my preferences.*

Acceptance Criteria:

- A feedback text area is displayed on the result page below the workout plan
- User enters textual feedback (minimum 3, maximum 1000 characters)
- System sends original plan and feedback text to Gemini Pro for revision
- A complete revised 7-day plan is generated and displayed to the user
- Updated plan is saved in the database; original plan is preserved
- A success message "Plan updated successfully based on your feedback!" is shown

US-04 | Admin – View All Users | Story Points: 3 | Priority: Medium

*As an **administrator**, I want to view all registered users and their generated plans in a single dashboard so that I can monitor usage and manage records.*

Acceptance Criteria:

- Admin dashboard is accessible at the /view-all-users route
- All users listed with: name, age, weight, goal, intensity, original plan, updated plan
- Shows "Not updated" if no feedback-based plan exists for a user
- Each user row has a delete button for removal

US-05 | Admin – Delete User | Story Points: 2 | Priority: Medium

*As an **administrator**, I want to delete a user and their workout plans so that I can remove test data or inactive accounts from the system.*

Acceptance Criteria:

- Delete button is available for each user on the admin dashboard
- On deletion, user record and all associated plans are removed from DB (cascade delete)
- After deletion, admin is redirected back to /view-all-users
- Deleted user no longer appears in the dashboard list

References:

1. [FastAPI Documentation – https://fastapi.tiangolo.com/](https://fastapi.tiangolo.com/)
2. [Google Gemini API – https://ai.google.dev/gemini-api/docs](https://ai.google.dev/gemini-api/docs)
3. [SQLAlchemy ORM – https://docs.sqlalchemy.org/en/20/](https://docs.sqlalchemy.org/en/20/)

4. [Pydantic v2](https://docs.pydantic.dev/latest/) – <https://docs.pydantic.dev/latest/>